LECTURE-6

Cell

Acknowledgements: Some of the materials are taken from v different sources in Google besides from Campbell's book on biology

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Overview

KEY CONCEPTS

6.1 Biologists use microscopes and the tools of biochemistry to study cells

6.2 Eukaryotic cells have internal membranes that compartmentalize their functions

6.3 The eukaryotic cell's genetic instructions are housed in the nucleus and carried out by the ribosomes

6.4 The endomembrane system regulates protein traffic and performs metabolic functions in the cell

6.5 Mitochondria and chloroplasts change energy from one form to another

6.6 The cytoskeleton is a network of fibers that organizes structures and activities in the cell

6.7 Extracellular components and connections between cells help coordinate cellular activities

Mitochondria and chloroplasts change energy from one form to another

- Mitochondria and chloroplasts are the **organelles that convert energy to forms that cells can use for work**.

- **Mitochondria** are the **sites of cellular respiration**, using oxygen to generate ATP by extracting energy from sugars, fats, and other fuels.

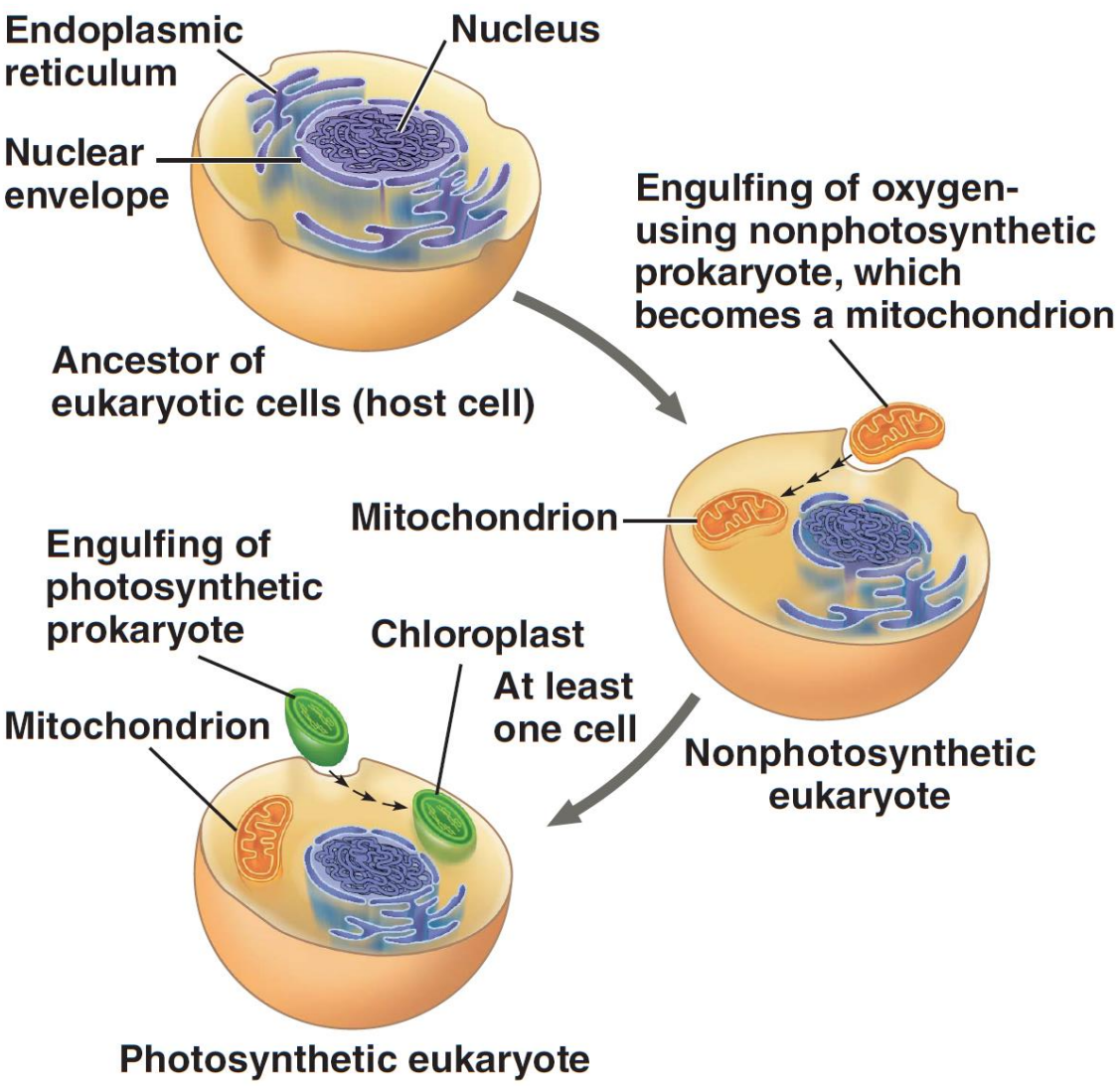
- **Chloroplasts**, found in plants and algae, are the **sites of photosynthesis**.

  - ✓ *Chloroplasts convert solar energy to chemical energy by absorbing sunshine and using it to synthesize new organic compounds such as sugars from CO_2 and H_2O.*

Mitochondria and chloroplasts have a similar evolutionary origin.

- The **endosymbiont theory** states that an early ancestor of eukaryotic cells engulfed an oxygen-using nonphotosynthetic prokaryotic cell.

  - ✓ *The engulfed cell became an endosymbiont within its host cell.*
  - ✓ *Over the course of evolution, the host cell and its endosymbiont merged into a single organism, a eukaryotic cell with a mitochondrion.*
  - ✓ *One of these cells engulfed a photosynthetic prokaryote and evolved into a eukaryotic cell containing chloroplasts.*



Endosymbiont theory

- There is considerable **evidence to support the endosymbiont theory** for the origin of mitochondria and chloroplasts.

  - ✓ *In contrast to organelles of the endomembrane system, each mitochondrion or chloroplast has two membranes separating the innermost space from the cytosol.*

  - ✓ *Mitochondria and chloroplasts contain ribosomes and circular DNA molecules that are attached to their inner membranes.*

  - ✓ *The DNA directs the synthesis of some of the organelle's proteins, which are made on the ribosomes inside the organelles.*

  - ✓ *Mitochondria and chloroplasts grow and reproduce as semiautonomous organelles.*

Mitochondria convert chemical energy within eukaryotic cells.

- Almost all eukaryotic cells have mitochondria.
  - *Cells may have one very large mitochondrion or hundreds to thousands of individual mitochondria.*
  - *The number of mitochondria is correlated with aerobic metabolic activity.*
  - *Mitochondria have a smooth outer membrane and a convoluted inner membrane with infoldings called **cristae**.*
- The inner membrane divides the mitochondrion into **two internal compartments**.
  - *The first compartment is the intermembrane space, a narrow region between the inner and outer membranes.*
  - *The inner membrane encloses the **mitochondrial matrix**, a fluid-filled space with mitochondrial DNA, ribosomes, and enzymes.*

Mitochondria–Structure

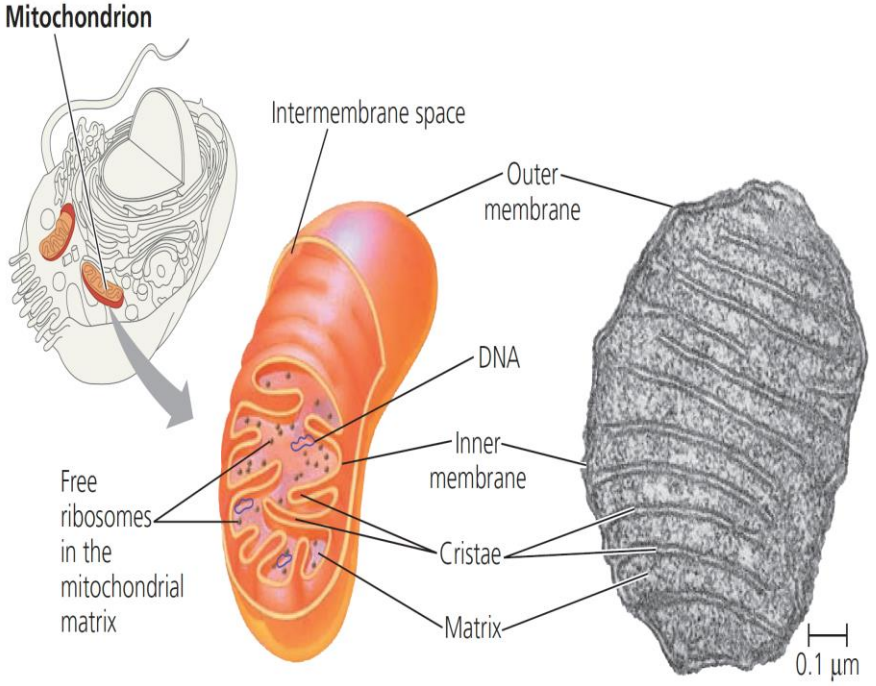
- Some of the metabolic steps of cellular respiration are catalyzed by enzymes in the matrix.

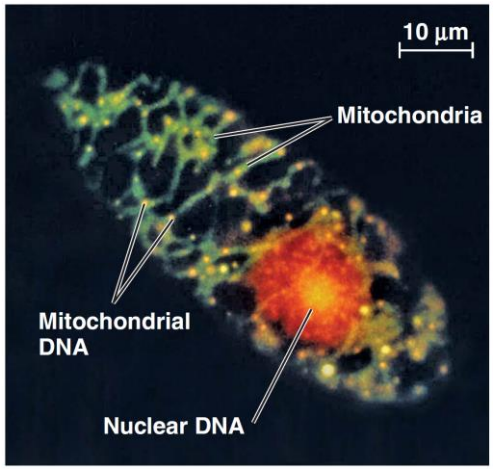
- Other proteins that function in respiration, including the enzyme that makes ATP, are built into the inner membrane.

  - ✓ *The highly folded cristae present a large surface area for these enzymes.*

- Mitochondria are about 1–10 µm long, in general.

  - ✓ *Mitochondria are dynamic: moving, changing shape, fusing, and dividing.*





Chloroplast:Capture light energy and convert it to chemical energy.

- Chloroplasts **contain the green pigment chlorophyll** as well as enzymes and other molecules that function in the photosynthetic production of sugar.
  - *They are 3–6 µm in diameter and are found in leaves and other green organs of plants and algae.*

- The contents of the chloroplast are separated from the cytosol by an envelope consisting of two membranes separated by a narrow intermembrane space.

- Inside the innermost membrane is a fluid-filled space, the **stroma**, in which float membranous sacs, the **thylakoids**.

  - *The stroma contains chloroplast DNA, ribosomes, and enzymes.*
  - *The thylakoids are flattened sacs that play a critical role in converting light to chemical energy. In some regions, thylakoids are stacked like poker chips into **grana**.*

Chloroplast

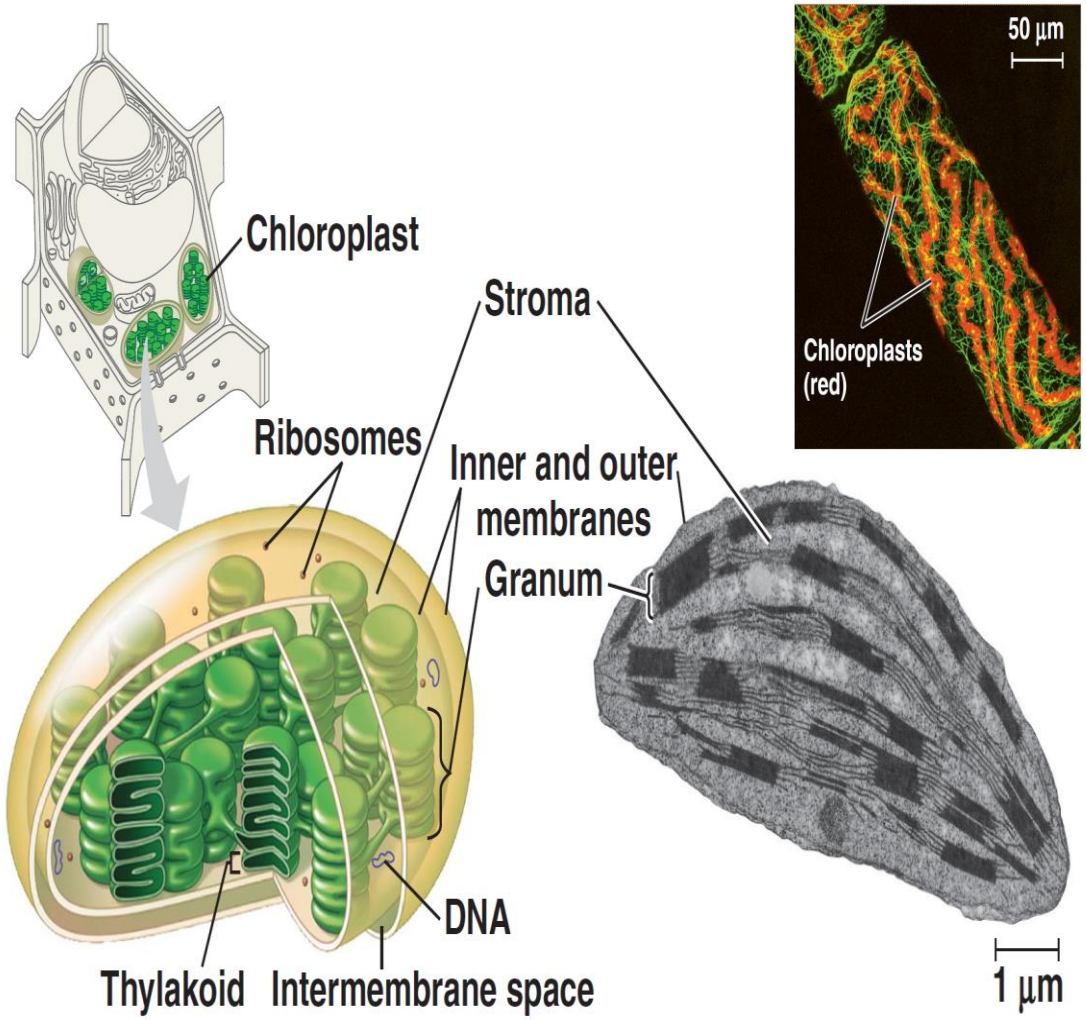
- The membranes of the chloroplast divide the chloroplast into **three compartments**: the intermembrane space, the stroma, and the thylakoid space.

- Like mitochondria, chloroplasts are dynamic structures.

  - ✓ *Their shape is plastic. They are mobile and move around the cell along tracks of the cytoskeleton.*

- The chloroplast belongs to **a family of plant structures called plastids**.

  - *Amyloplasts* are *colorless plastids that store starch in roots and tubers*.
  - *Chromoplasts* store *pigments for fruits and flowers*.

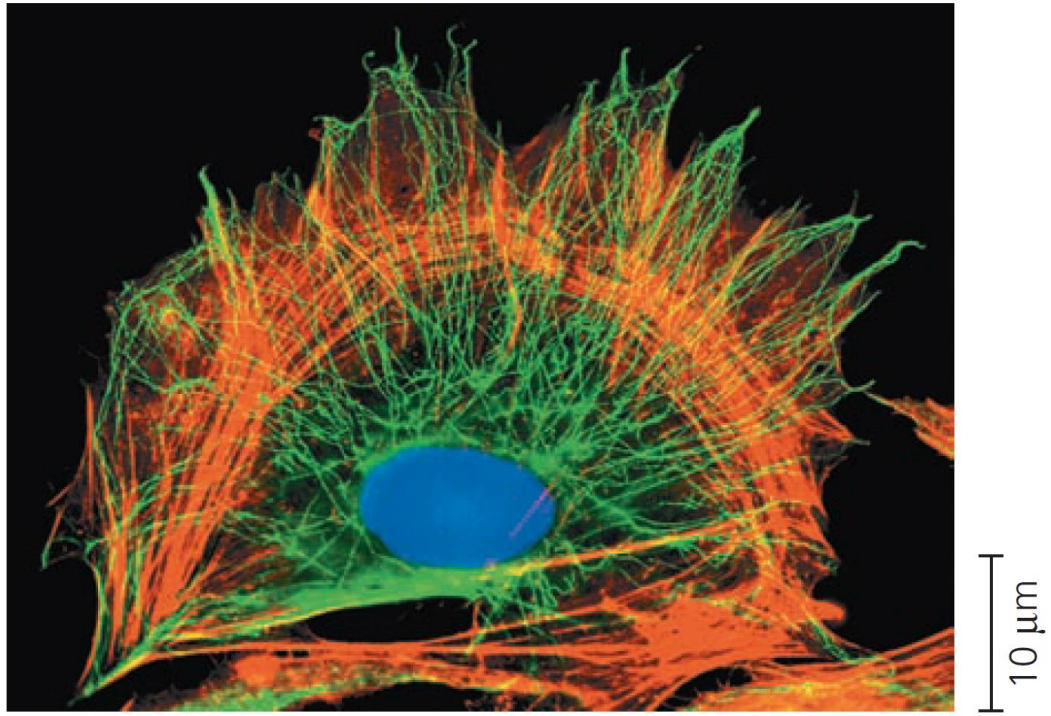


Cytoskeleton:provides support, motility, and regulation.

- The **cytoskeleton** is **a network of fibers extending through the cytoplasm that provides mechanical support and maintains the cell's shape**.

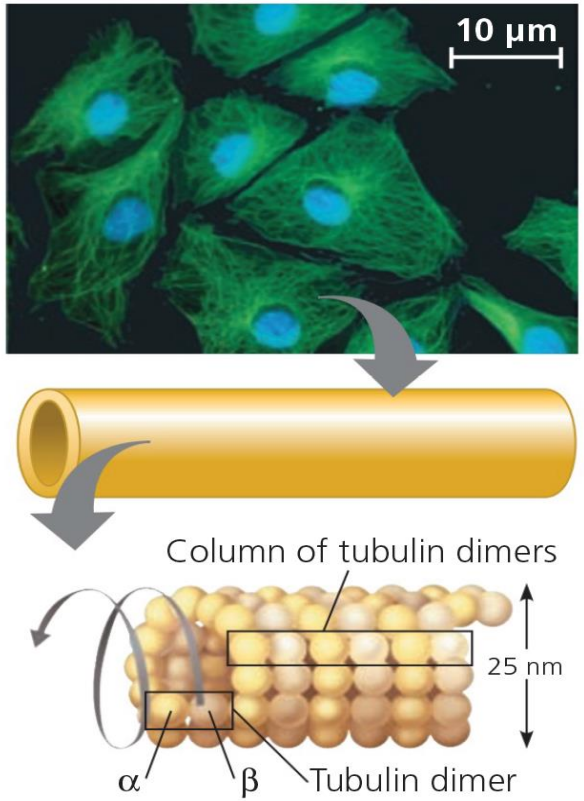
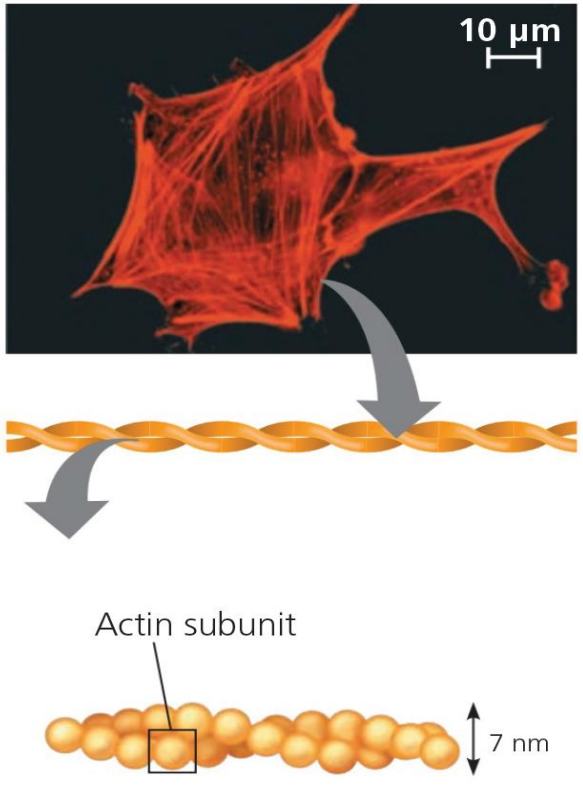
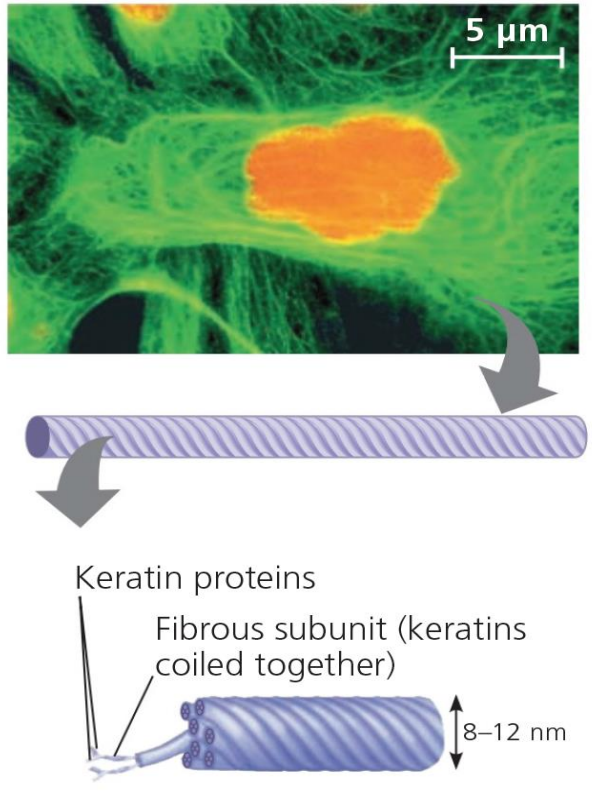
  - ✓ *This function is especially important in animal cells, which lack walls*.

- The cytoskeleton **provides anchorage** for many organelles and cytosolic enzymes.
- The cytoskeleton is **dynamic** and can be dismantled in one part and reassembled in another to change the shape of the cell.



▲ **Figure 6.20 The cytoskeleton.** As shown in this fluorescence micrograph, the cytoskeleton extends throughout the cell. The cytoskeletal elements have been tagged with different fluorescent molecules: green for microtubules and reddish orange for microfilaments. A third component of the cytoskeleton, intermediate filaments, is not evident here. (The blue color tags the DNA in the nucleus.)

Table 6.1 The Structure and Function of the Cytoskeleton

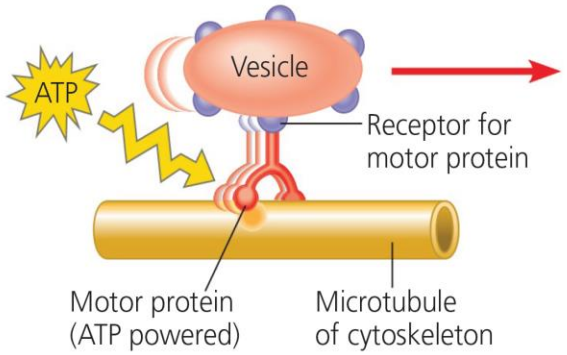
Property	Microtubules (Tubulin Polymers)	Microfilaments (Actin Filaments)	Intermediate Filaments
Structure	Hollow tubes	Two intertwined strands of actin	Fibrous proteins coiled into cables
Diameter	25 nm with 15-nm lumen	7 nm	8–12 nm
Protein subunits	Tubulin, a dimer consisting of α-tubulin and β-tubulin	Actin	One of several different proteins (such as keratins)
Main functions	Maintenance of cell shape (compression-resisting "girders"); cell motility (as in cilia or flagella); chromosome movements in cell division; organelle movements	Maintenance of cell shape (tension-bearing elements); changes in cell shape; muscle contraction; cytoplasmic streaming in plant cells; cell motility (as in amoeboid movement); division of animal cells	Maintenance of cell shape (tension-bearing elements); anchorage of nucleus and certain other organelles; formation of nuclear lamina
Fluorescence micrographs of fibroblasts. Fibroblasts are a favorite cell type for cell biology studies. In each, the structure of interest has been tagged with fluorescent molecules. The DNA in the nucleus has also been tagged in the first micrograph (blue) and third micrograph (orange).			

Vesicle transport: Microtubule and motor protein

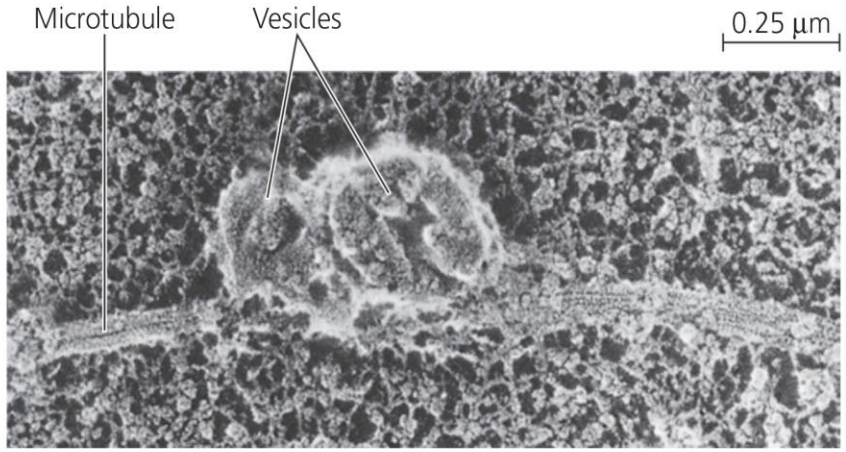
- **Microtubules shape and support the cell and serve as tracks to guide motor proteins carrying organelles to their destination.**

  - *Microtubules guide secretory vesicles from the Golgi apparatus to the plasma membrane.*

  - *Microtubules are also responsible for the separation of chromosomes during cell division.*



(a) Motor proteins that attach to receptors on vesicles can "walk" the vesicles along microtubules or, in some cases, along microfilaments.



(b) In this SEM of a squid giant axon (a nerve cell extension), two vesicles containing neurotransmitters move toward the axon's tip.

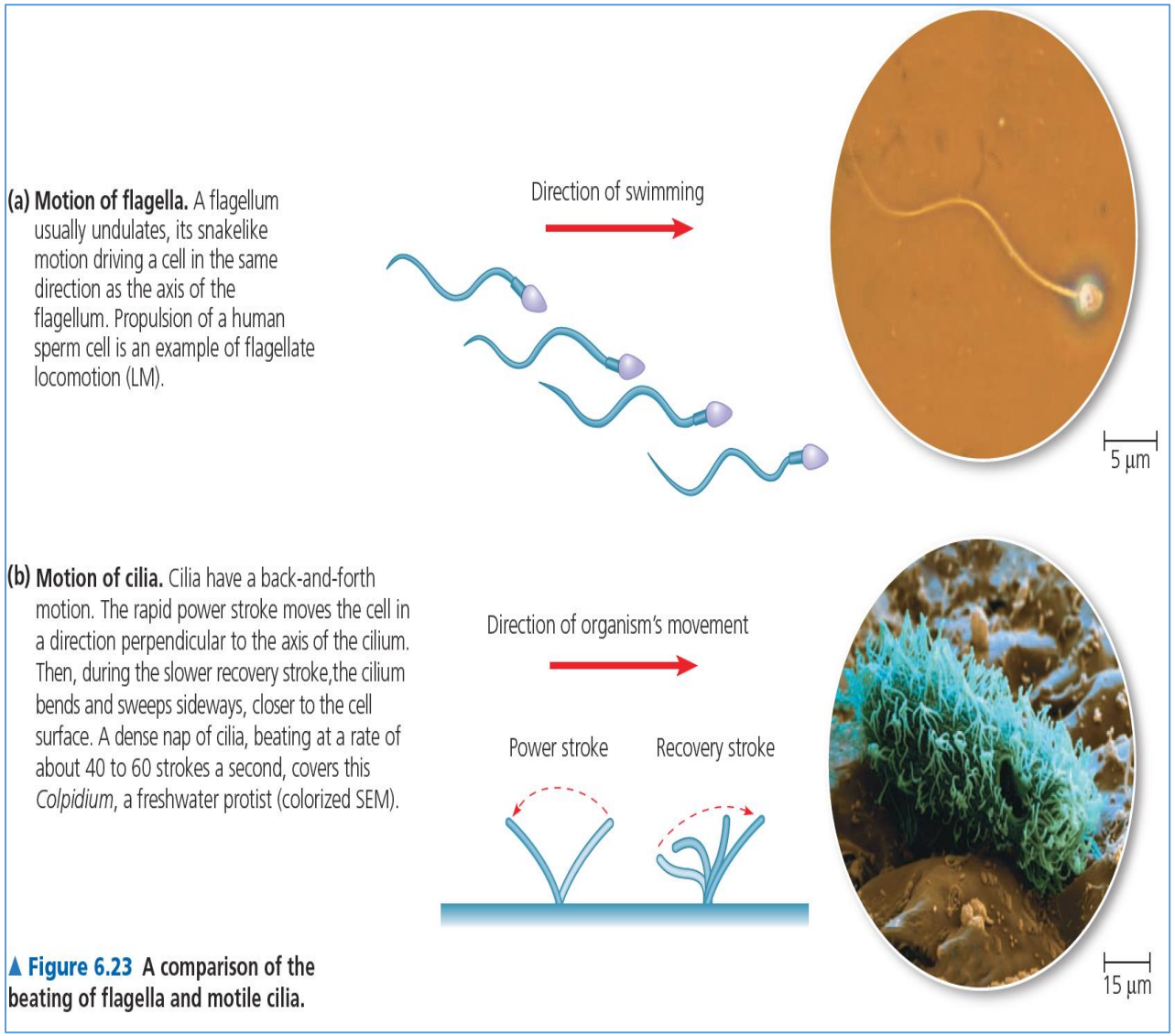
▲ **Figure 6.21 Motor proteins and the cytoskeleton.**

Microtubules- cilia and flagella for locomotion

- In eukaryotes, a specialized arrangement of microtubules is responsible for the beating of flagella (singular, flagellum) and cilia (singular, cilium), microtubule-containing extensions that project from some cells.

- Many unicellular eukaryotes are propelled through water by cilia or flagella that act as locomotor appendages, and the sperm of animals, algae, and some plants have flagella.

- Motile cilia usually occur in large numbers on the cell surface.



▲ **Figure 6.23** A comparison of the beating of flagella and motile cilia.

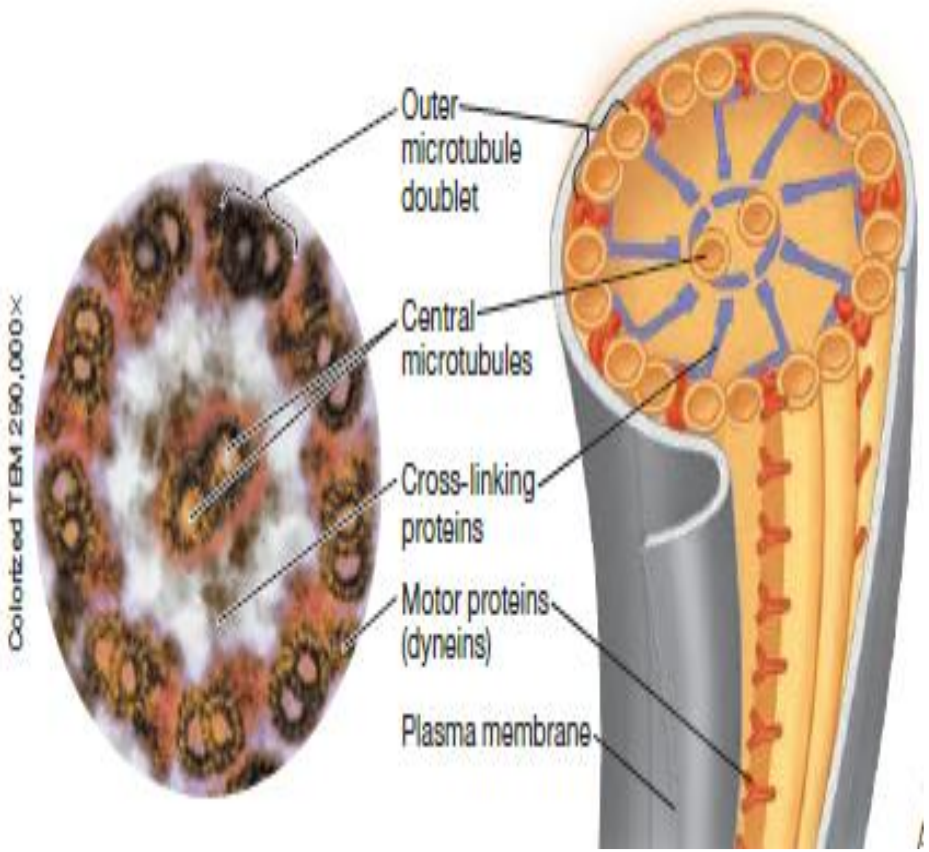
Structure of microtubules

- Though different in length and beating pattern, cilia and flagella have a common structure and mechanism of movement.

- Both are composed of microtubules wrapped in an extension of the plasma membrane.

- In nearly all eukaryotic cilia and flagella, nine microtubule doublets surround a central pair of microtubules. This arrangement is called the “9 + 2” pattern.

- The microtubule assembly is anchored in the cell by a basal body (not shown in the figure), structurally similar to a centriole.



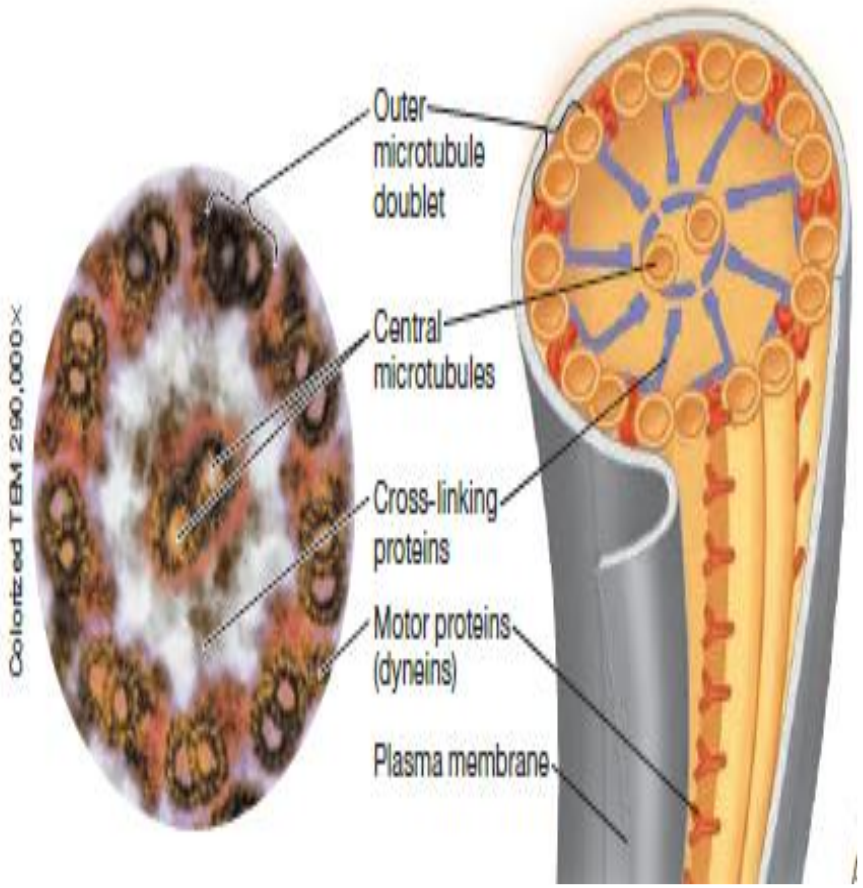
How microtubules brings movement in the cell?

- Large motor proteins called dyneins (red in the figure) are attached along each outer microtubule doublet.

- A dynein protein has two “feet” that “walk” along an adjacent doublet. The walking movement is coordinated to happen on one side at a time.

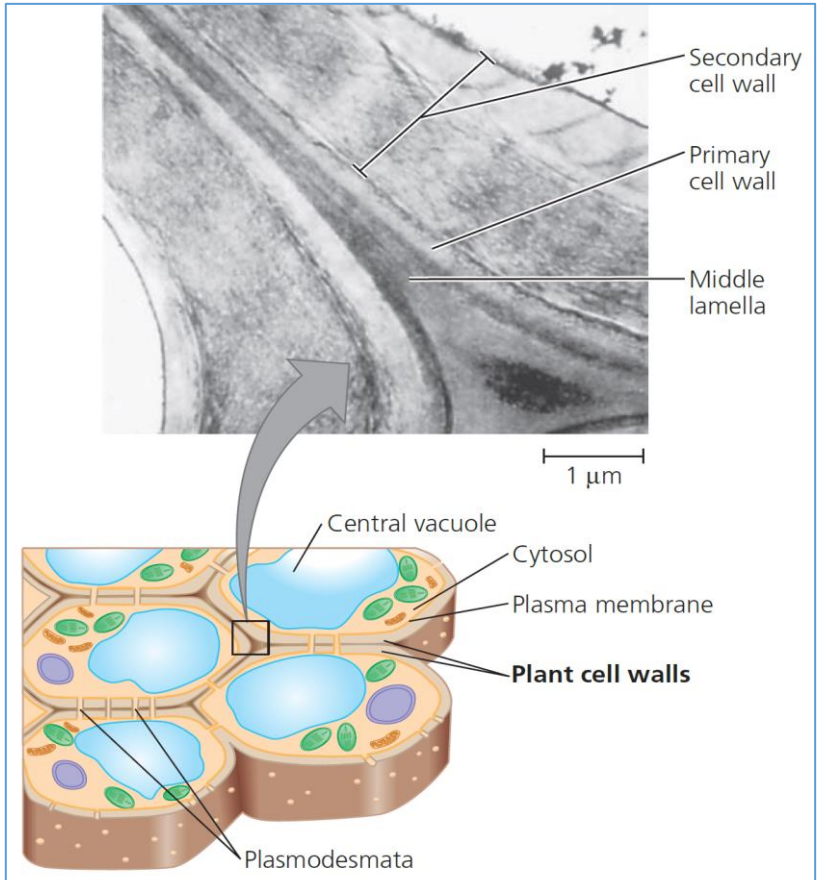
- The microtubules are held together by flexible cross-linking proteins (purple in the diagram). If the doublets were not held, they would slide past each other.

- Instead, the “walking” of the dynein feet causes the microtubules—and consequently, the cilium or flagellum—to bend.

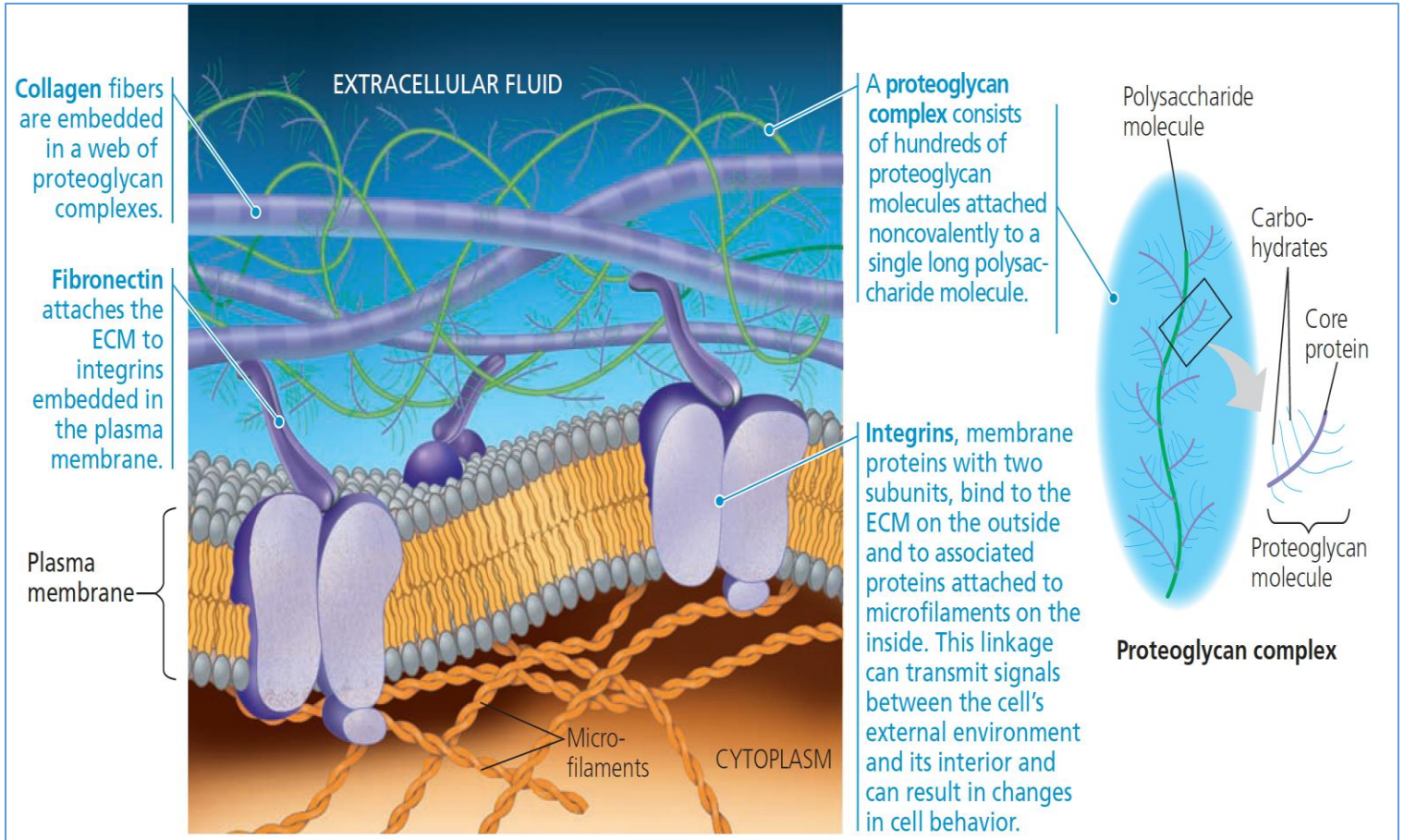


Extracellular matrix

❖The extracellular matrix of animal cells provides support, adhesion, movement, and regulation.



▲ **Figure 6.27 Plant cell walls.** The drawing shows several cells, each with a large vacuole, a nucleus, and several chloroplasts and mitochondria. The transmission electron micrograph shows the cell walls where two cells come together. The multilayered partition between plant cells consists of adjoining walls individually secreted by the cells.



▲ **Figure 6.28 Extracellular matrix (ECM) of an animal cell.** The molecular composition and structure of the ECM vary from one cell type to another. In this example, three different types of ECM molecules are present: proteoglycans, collagen, and fibronectin.

ECM

- Though lacking cell walls, animal cells do have an elaborate **extracellular matrix (ECM)**.

- The primary constituents of the ECM are glycoproteins, especially **collagen** fibers, embedded in a network of glycoprotein **proteoglycans**.

  - **Collagen accounts for about half the total protein in the human body**.

  - A **proteoglycan molecule** consists of a small core protein with many carbohydrate chains covalently attached, so that it may be up to 95% carbohydrate.

  - Large proteoglycan complexes can form when hundreds of proteoglycans become noncovalently attached to a single long polysaccharide molecule.

ECM – Fibronectins and integrins

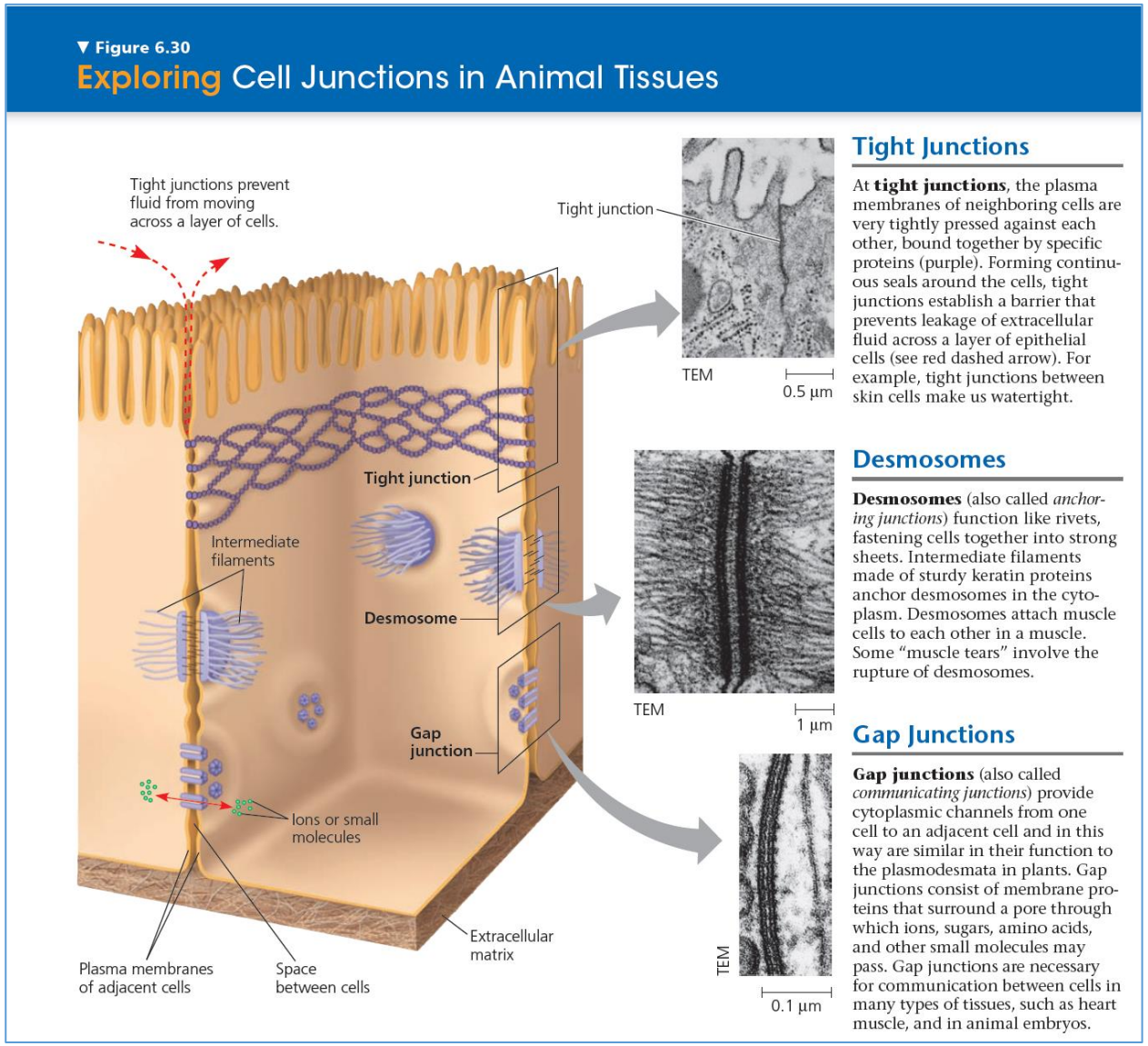
- In many cells, **fibronectins** in the ECM connect to **integrins,** cell-surface receptor proteins that span the membrane and bind on their cytoplasmic side to proteins attached to microfilaments of the cytoskeleton.

- The interconnections from the ECM to the cytoskeleton via the fibronectin-integrin link permit the integration of changes inside and outside the cell.

- Mechanical signaling involves fibronectin, integrins, and microfilaments of the cytoskeleton.

  - ✓ *Changes in the cytoskeleton may in turn trigger chemical signaling pathways inside the cell, leading to changes in the set of proteins being made by the cell and thus changes in the cell's function.*
  - ✓ *This may coordinate the behavior of all the cells within a tissue.*

ECM- Junctions for communication



- Neighboring cells in tissues, organs, and organ systems often adhere, interact, and communicate through direct physical contact.
- Plant cells are perforated with **plasmodesmata**, channels allowing cytosol to pass between cells.
  - ✓ Water and small solutes can pass freely from cell to cell.
  - ✓ In certain circumstances, proteins and RNA can be exchanged.
- Macromolecules being transported to neighboring cells reach the plasmodesmata by moving along fibers of the cytoskeleton

ECM – Junctions

- Animals have three main types of intercellular links**:**
  - ✓ ***Tight junctions*, *desmosomes*, and *gap junctions*.**
- In **tight junctions,** membranes of adjacent cells are fused, forming continuous belts around cells that prevent leakage of extracellular fluid.

- **Desmosomes** (or anchoring junctions) fasten cells together into strong sheets, much like rivets.
  - Intermediate filaments of keratin reinforce desmosomes.

- **Gap junctions** (or communicating junctions) provide cytoplasmic channels between adjacent cells.
  - ✓ *Special membrane proteins surround these pores.*
  - ✓ *Ions, sugars, amino acids, and other small molecules can pass.*
  - ✓ *In embryos, gap junctions facilitate chemical communication during development.*

Extra slides

Extras

Microtubules: Classification

- ***Three main types of fibers make up the cytoskeleton: microtubules, microfilaments, and intermediate filaments.***

- ***Microtubules*** **are the thickest** of the three types of fibers; *microfilaments* (or actin filaments) are the thinnest; and *intermediate filaments* are fibers with diameters in a middle range.

- **Microtubules** are **hollow rods** about 25 nm in diameter and 200 nm to 25 µm in length.

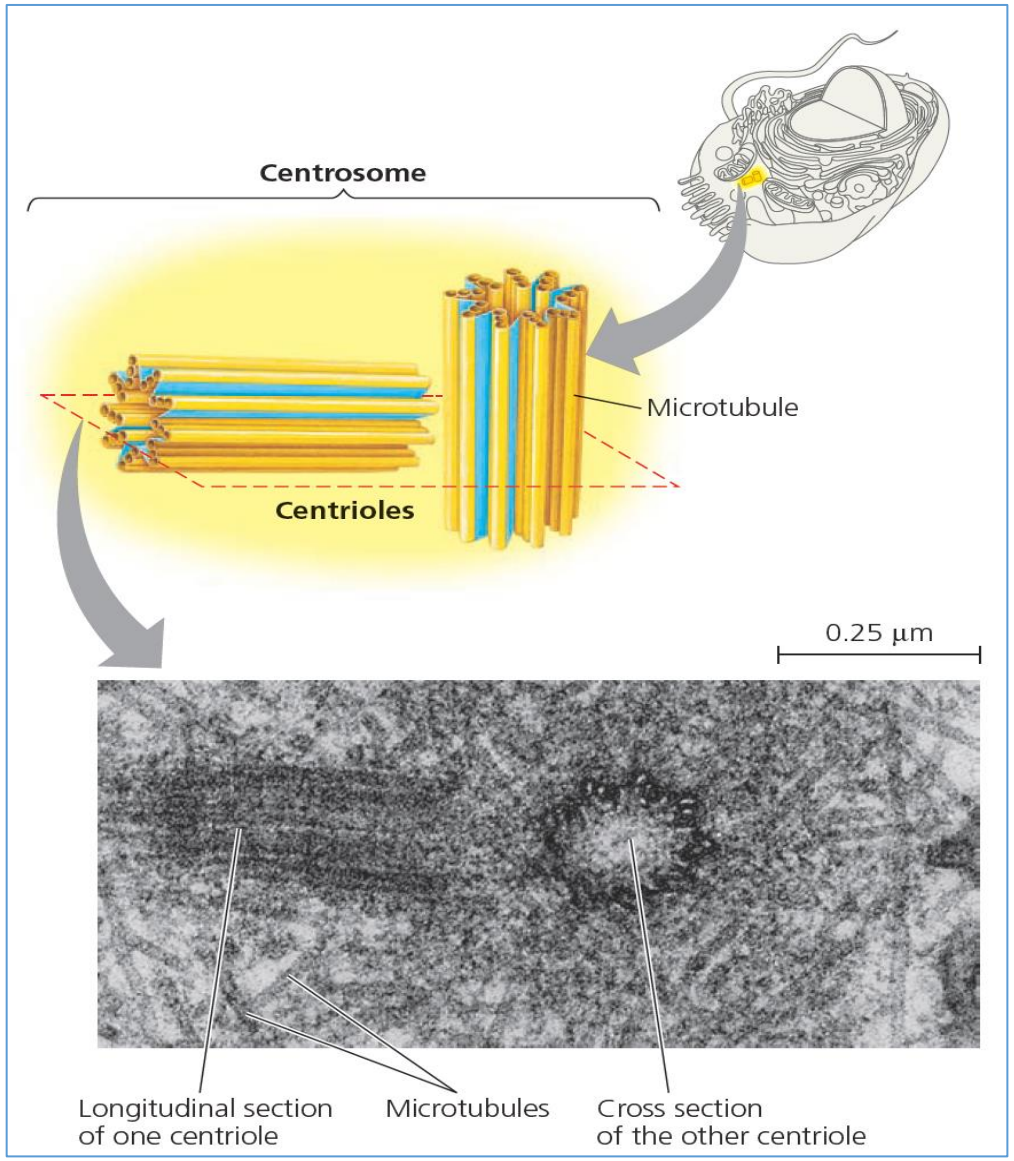
- Microtubule fibers are **constructed of the globular protein tubulin**.

  - ✓ *Each tubulin molecule is a dimer consisting of two subunits.*
  - ✓ *A tubulin dimer consists of two slightly different polypeptides: α-tubulin and β-tubulin*

Microtubule: Centrosome and centriole

- A microtubule changes in length by adding or removing tubulin dimers.
  - ✓ *The two ends of a microtubule are slightly different.*
  - ✓ *The "plus" end can accumulate or release tubulin dimers at a much higher rate than the other end.*

- **Microtubules shape and support the cell and serve as tracks to guide motor proteins carrying organelles to their destination.**
  - ✓ *Microtubules guide secretory vesicles from the Golgi apparatus to the plasma membrane.*
  - ✓ *Microtubules are also responsible for the separation of chromosomes during cell division.*



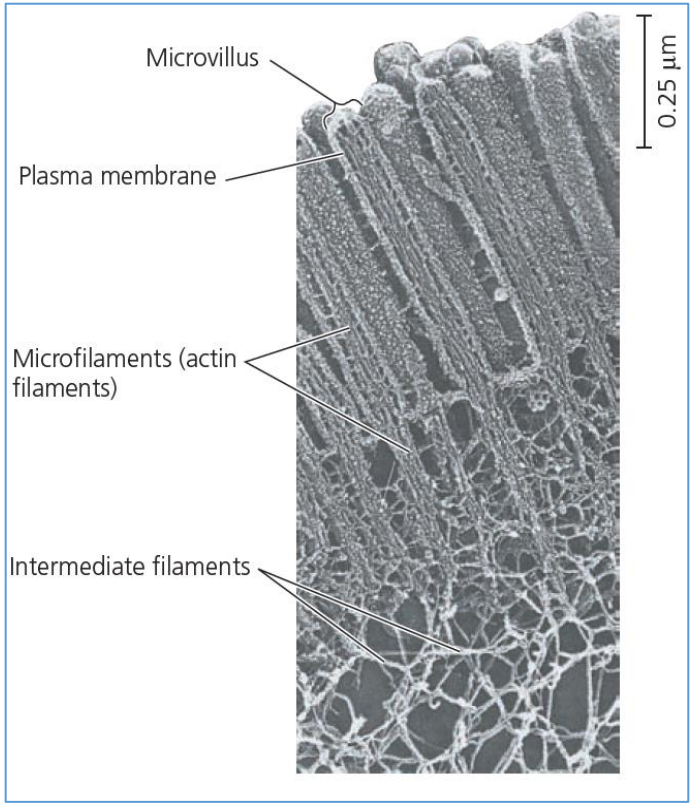
Microfilaments – structure

- **<u>Microfilaments</u>** or **actin filaments** are solid rods about 7 nm in diameter, present in all eukaryotic cells.

- Each microfilament is built as a twisted double chain of actin subunits.

- Microfilaments **can form structural networks because of their ability to branch**.

- The structural role of microfilaments in the cytoskeleton is to bear tension, resisting pulling forces within the cell.



- *Cortical microfilaments* form a three-dimensional network just inside the plasma membrane to help support the cell's shape, giving the cell **cortex** the semisolid consistency of a gel.

Intermediate filaments

- **<u>Intermediate filaments</u>** range in diameter from 8 to 12 nm, larger than microfilaments but smaller than microtubules.

- Intermediate filaments are a diverse class of cytoskeletal units, built from a family of proteins that includes the keratins.

- Like microfilaments, intermediate filaments are specialized for bearing tension.

- **Intermediate filaments are more permanent fixtures of the cytoskeleton than are the other two classes.**

  - ✓ *Even after cells die, intermediate filament networks often persist.*
  - ✓ *The networks reinforce cell shape and fix organelle location*

Intermediate filaments

- **The nucleus sits within a cage made of intermediate filaments**, fixed in location by branches of the filaments that extend into the cytoplasm.
  - ✓ *Other intermediate filaments make up the nuclear lamina that lines the interior of the nuclear envelope.*
- When the shape of the entire cell is correlated with function, intermediate filaments support that shape.
  - ✓ *The axons of nerve cells are strengthened by one class of intermediate filament.*
- Various kinds of intermediate filaments function as the framework of the entire cytoskeleton

Cytoskeleton: role in motility

- The cytoskeleton also **plays a major role in *cell motility***, including changes in cell location and limited movements of parts of the cell.

- The cytoskeleton interacts with **motor proteins** to produce motility.

  - ✓ *Cytoskeleton elements and motor proteins work together with plasma membrane molecules to move the whole cell along fibers outside the cell.*

  - ✓ *Motor proteins bring about movements of **cilia and flagella** by gripping cytoskeletal components such as microtubules and moving them past each other.*

  - ✓ *A similar mechanism causes **muscle cells** to contract.*

Cytoskeleton–vesicles use motor protein

- Inside the cell, **vesicles use motor protein "feet" to "walk" to destinations along a track provided by the cytoskeleton**.
  - ✓ *This is how vesicles containing neurotransmitter molecules migrate to the tips of axons.*
- The **vesicles** that bud off from the ER travel to the Golgi apparatus along tracks built of cytoskeletal elements.
- The cytoskeleton manipulates the plasma membrane to form food vacuoles during phagocytosis.
- Cytoplasmic streaming in plant cells is caused by the cytoskeleton.

Cytoskeleton-biochemical activities

- The cytoskeleton also **plays a role in the regulation of biochemical activities** in the cell in response to mechanical stimulation.

  - ✓ *Cytoskeletal elements transmit forces exerted by extracellular molecules on cell surface proteins into the cell—and even into the nucleus.*

  - ✓ *In an experiment, investigators used a micromanipulation device to pull on certain plasma membrane proteins attached to the cytoskeleton, leading to almost instantaneous rearrangements of nucleoli and other structures in the nucleus.*

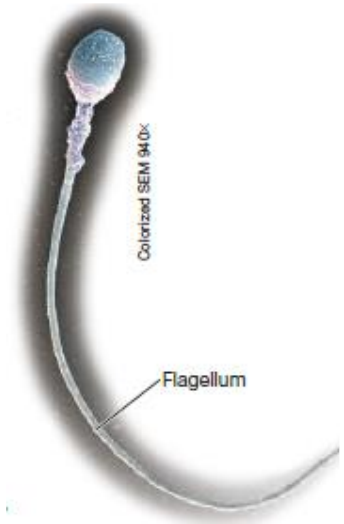
  - ✓ *In this way, cytoskeletal transmission of naturally occurring mechanical signals may help regulate and coordinate the cell's response.*

Internal structure of cilia or flagellum

- The role of the cytoskeleton in movement is seen in the motile appendages that protrude from certain cells.

- The short, numerous appendages that propel *Paramecium* are called **cilia** (singular, *cilium*).

- Other protists may move using flagella, longer than cilia and usually limited to one or a few per cell.



- These cilia sweep mucus containing trapped debris out of your lungs. (This cleaning function is impaired by cigarette smoke, which paralyzes the cilia.)

- A flagellum propels the cell by an undulating whiplike motion. In contrast, cilia work more like the coordinated oars of a rowing team